Swati Aggrawal

swatiagg357@gmail.com | github.com/swatiaggrawal |linkedin.com/in/swatiaggrawal

PROFESSIONAL SUMMARY

M.Sc. Computer Science graduate from Sapienza University of Rome specializing in Artificial Intelligence and Machine Learning. Experienced in developing AI and computer vision solutions using Python, PyTorch, TensorFlow, FastAPI, Docker, and AWS. Conducted research on Vision Transformers and continual learning, with an IEEE publication and hands-on experience building scalable machine learning applications. Passionate about applying AI to solve real-world industrial challenges.

SKILLS

- **Programming Languages:** Python, C++, Java, SQL, Bash
- **AI & Machine Learning:** Machine Learning, Deep Learning, Computer Vision, Vision Transformers, Continual Learning, Transfer Learning, PyTorch, TensorFlow, Scikit-learn, Keras, Hugging Face Transformers, Generative AI
- **Software Engineering:** Data Structures & Algorithms, Object-Oriented Programming (OOP), FastAPI, REST APIs, Git
- **Cloud & DevOps:** Docker, AWS (ECS, Fargate, CloudWatch), Linux
- **Data Science & Visualization:** NumPy, Pandas, Matplotlib

EDUCATION

Sapienza Università di Roma **May 2026**

Master of Science in Computer Science

- **Relevant Coursework:** Natural Language Processing, Deep Learning and Applied AI, Machine Learning, Computer Vision, Distributed Systems
- **Thesis:** Evaluating Pretrained Vision Transformers Under Continual Learning.
- Developed and evaluated adapter-based continual learning methods for Vision Transformers using PyTorch and Elastic Weight Consolidation (EWC) to mitigate catastrophic forgetting.

Sharda University **July 2023**

Bachelor of Technology - CSE with AI and ML

- **Relevant Coursework:** Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- **Thesis:** Virtual Generative Model Inducing Human Identity to Computers in Metaverse
- Awarded **merit-based scholarship** (Top 5–10% of cohort).

EXPERIENCE

Researcher — Vision Lab, Sapienza Università di Roma (Master's Thesis)

Nov 2025 - May 2026

- Conducted research under Prof. Marco Raoul Marini on continual learning for computer vision in the Vision Lab.
- Evaluated pre-trained Vision Transformers on sequential image classification benchmarks.
- Designed and implemented adapter-based learning and Elastic Weight Consolidation (EWC) techniques to mitigate catastrophic forgetting.
- Developed PyTorch training and evaluation pipelines to benchmark continual learning methods using accuracy, forgetting, and computational efficiency metrics.
- Collaborated with researchers to document findings and compare state-of-the-art continual learning approaches.
- **Technologies:** Python, PyTorch, Vision Transformers, Computer Vision

PUBLICATION

Machine Learning for Air Pollution – IEEE Conference Publication (2023)

- Evaluated Random Forest, SVM, Decision Tree, and Neural Network models on environmental datasets.
- Performed comparative analysis using accuracy, robustness, and generalization metrics.
- Authored and presented findings through an IEEE peer-reviewed publication.

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PROJECTS

DocDelta AI-Powered Document Intelligence & Comparison Platform

- Developed an end-to-end AI document comparison platform using FastAPI and Sentence Transformers.
- Designed scalable embedding and retrieval pipelines for semantic document comparison.
- Containerized the application with Docker and deployed on AWS ECS/Fargate with CloudWatch monitoring.
- Built REST APIs and optimized API response time and inference latency for scalable deployment.
- **Technologies:** Python, FastAPI, Sentence Transformers, Docker, AWS ECS Fargate, CloudWatch, REST APIs

Metal Defect Detection using Computer Vision

- Developed computer vision models for automated industrial metal defect detection using CNNs, SVM, and Random Forest.
- Built image preprocessing, training, and evaluation pipelines for defect classification.
- Compared deep learning and traditional machine learning approaches to improve detection performance.
- Evaluated models using accuracy, confusion matrices, precision, recall, and F1-score.
- **Technologies:** Python, TensorFlow, Keras, OpenCV, Scikit-Learn

Lossy Text Compressor [Semantic NLP Pipeline]

- Developed a semantic text compression pipeline using transformer embeddings, vector quantization, and reconstruction techniques.
- Evaluated reconstruction quality using cosine similarity and ROUGE metrics.
- Analyzed trade-offs between compression efficiency and semantic preservation across multiple quantization levels.
- **Technologies:** Python, PyTorch, Transformer Models, NLP, Sentence Embeddings